

Problem Statement Title - Embedded Intelligent Microscopy System

PS Category - Hardware

Team Name (Registered on portal)- Team Gradient



Overview

- Manual counting and identification are time-consuming, error-prone, and subjective, creating bottlenecks in large-scale ecological studies.
- There is a critical need for automation using AI-enabled, embedded microscopy systems that can provide accurate, rapid, and on-site analysis..
- A low-cost, embedded solution with lightweight deep learning models will enable automatic identification, classification, and counting of plankton species

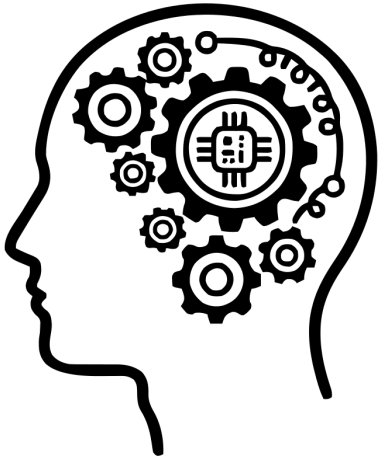
Features

LIGHTWEIGHT MODELS

Deep learning models optimized for embedded hardware, ensuring efficient performance even on low-resource devices like Raspberry Pi.

Automatic Identification

AI-enabled system for accurate species/genus-level classification of microscopic marine organisms directly from captured images.



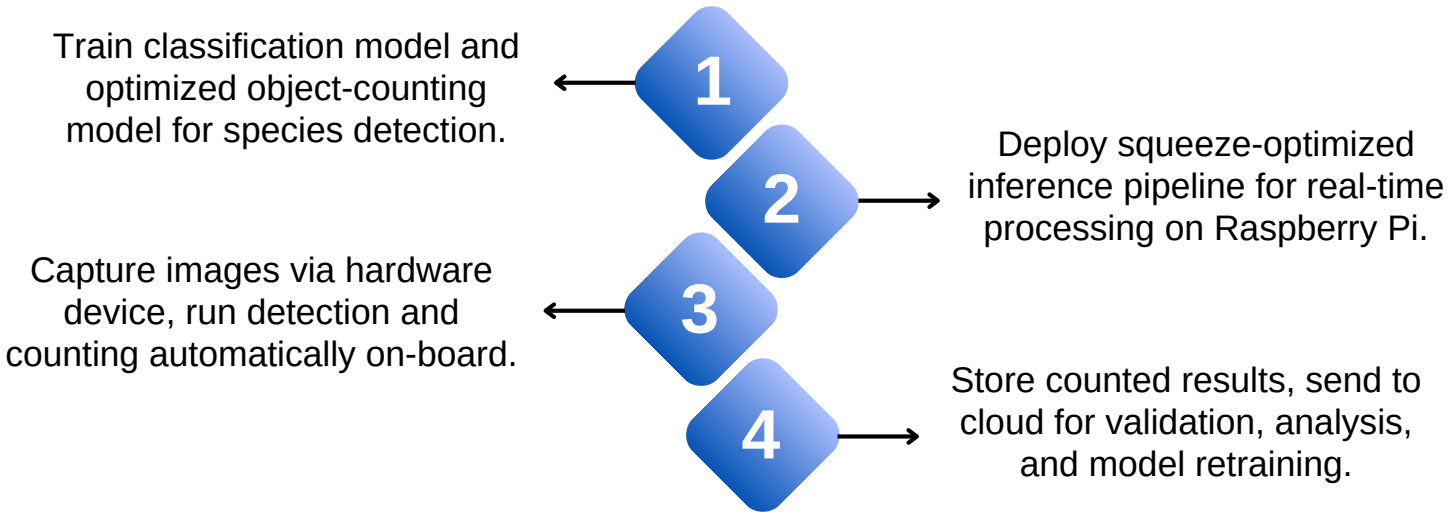
ACCURATE COUNTING

Reliable detection and enumeration of multiple organisms per frame, minimizing human error and speeding up biodiversity assessments.

ONBOARD PROCESSING

Integrated capability to process, store, and report identification results locally, enabling real-time ecological monitoring without external dependency.

Roadmap



Why Is Our Solution Innovative & Effective?

Lightweight & Efficient

Optimized for Raspberry Pi, ensuring low power, high-speed computation with minimal resources.

Self-Improving System

Error data is sent to the cloud and used for continuous retraining, making the model smarter over time.

Quality Assurance Check

Incorrect results are flagged and stored for later human verification, ensuring continuous accuracy.

Dual-Pipeline Optimization

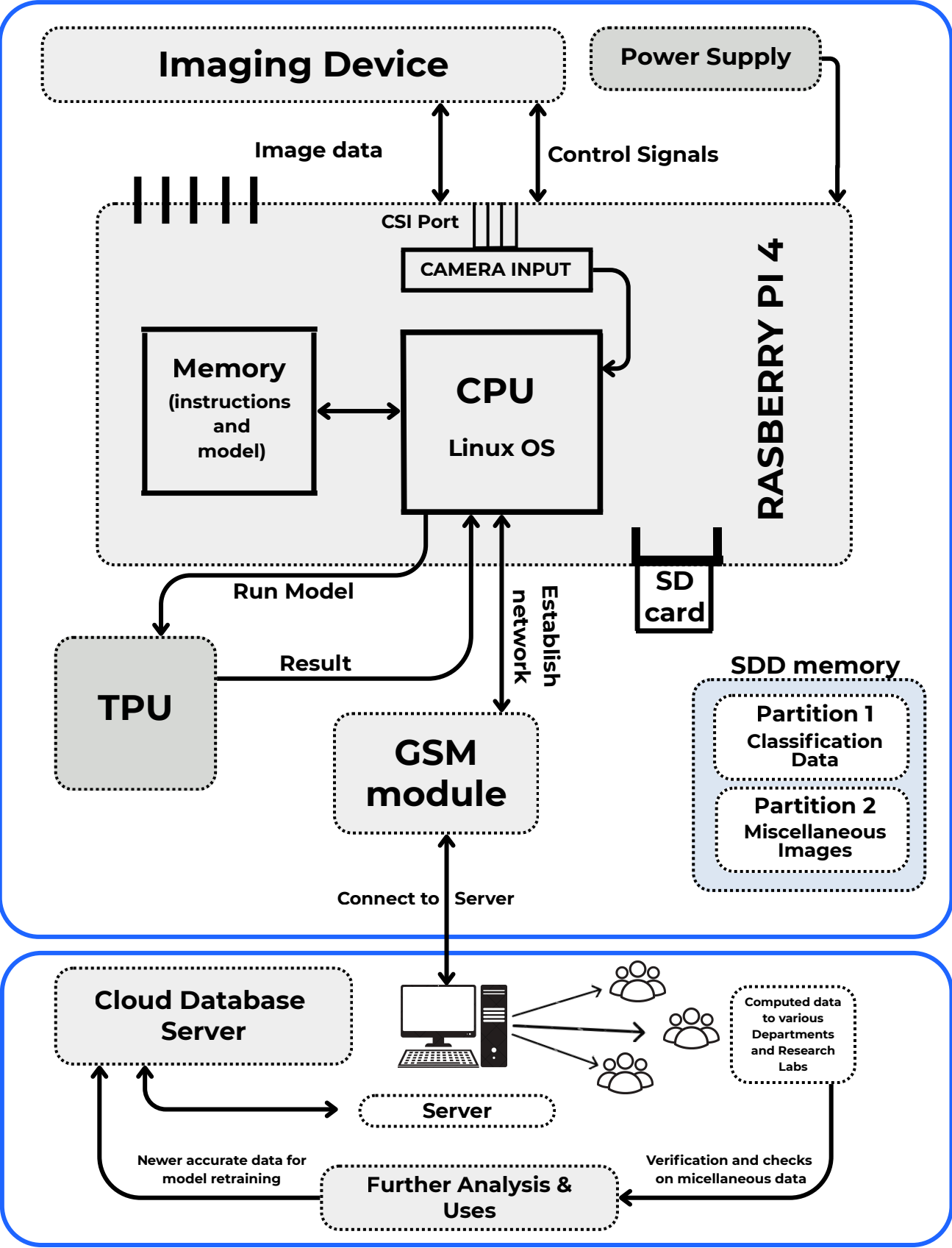
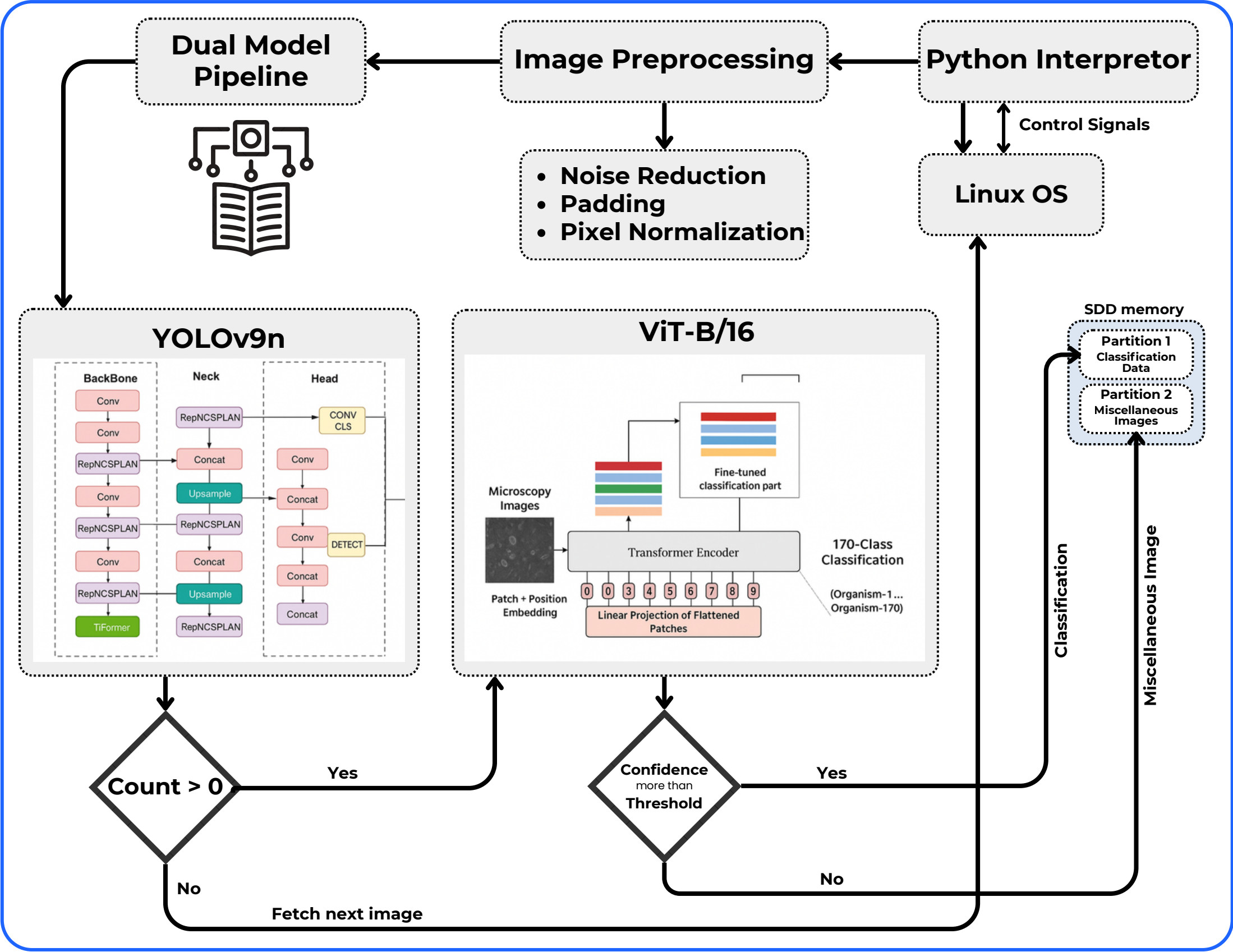
Filters presence first, then applies heavy models only when needed.

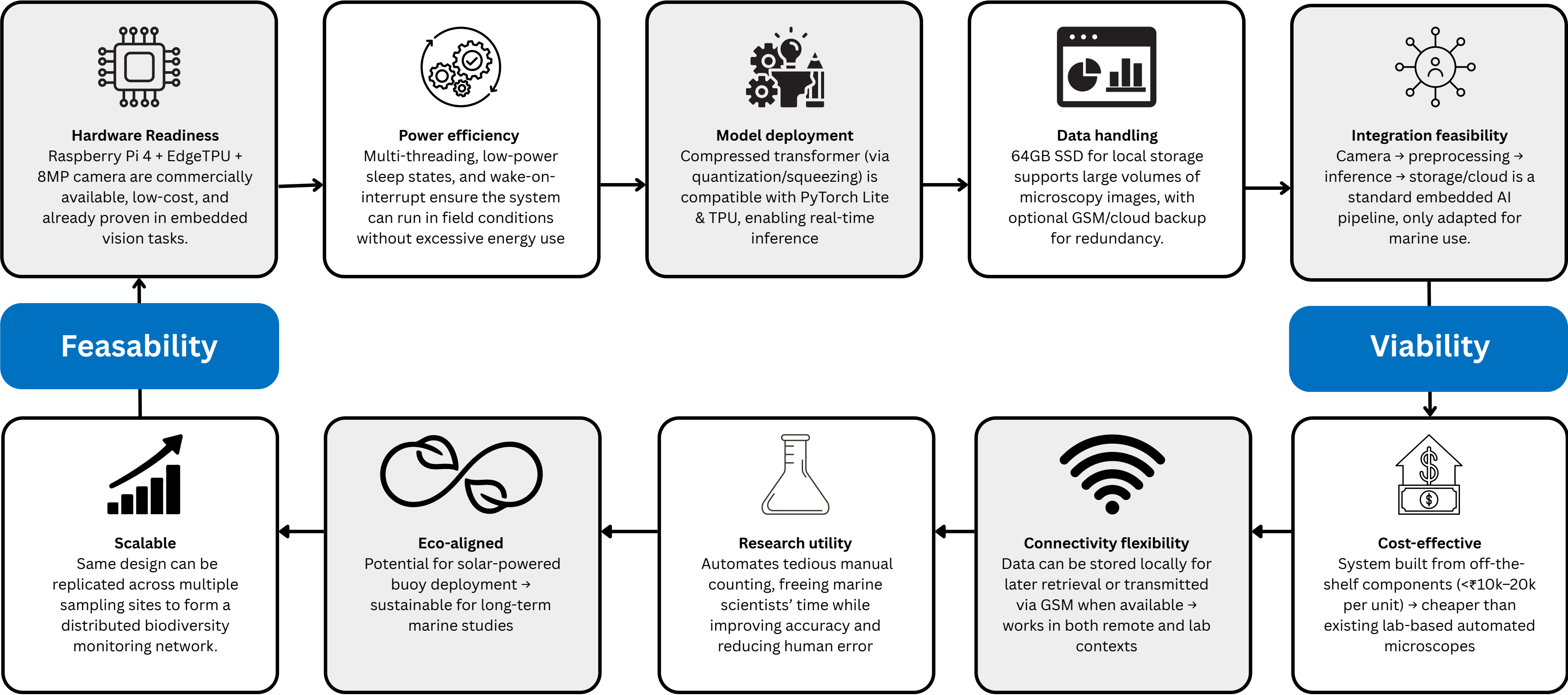
Cost-Effective Solution

Hardware implementation under ₹5K, making it scalable and affordable.

Seamless Connectivity

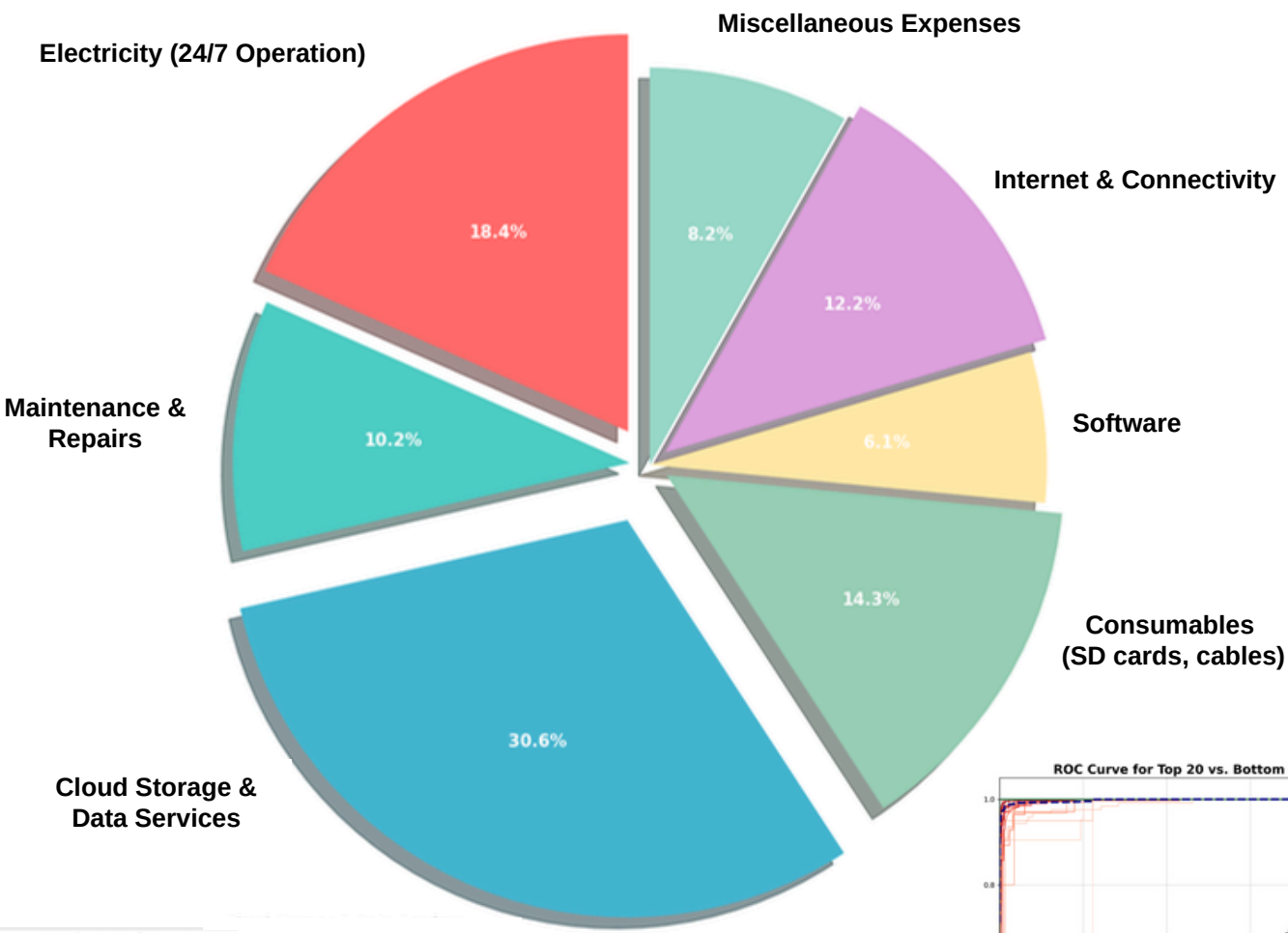
Wi-Fi-enabled device allows real-time monitoring, updates, and cloud integration.



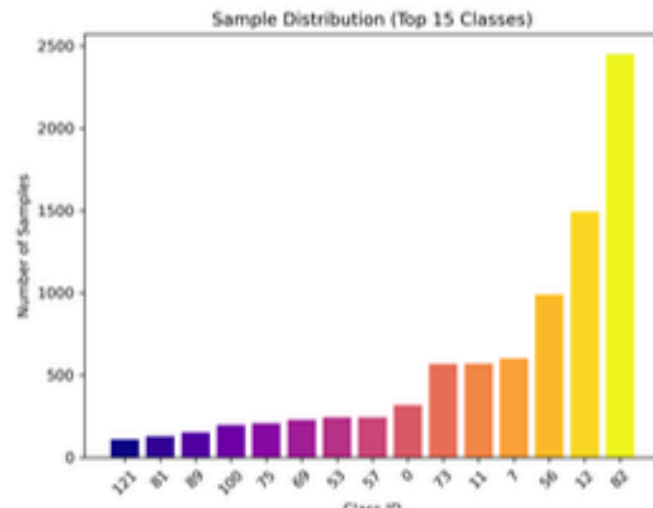
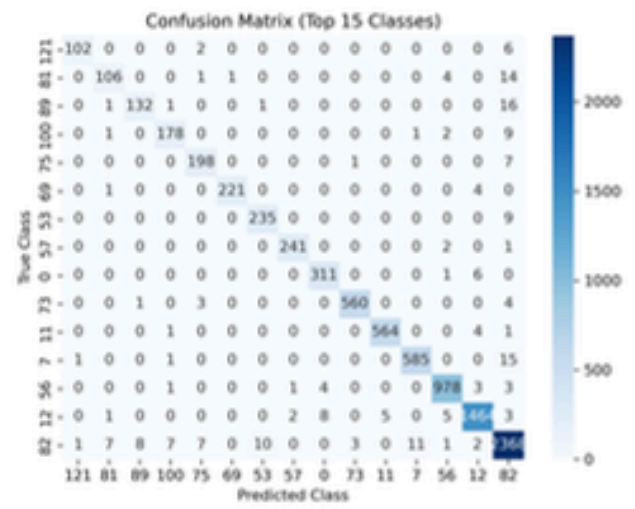
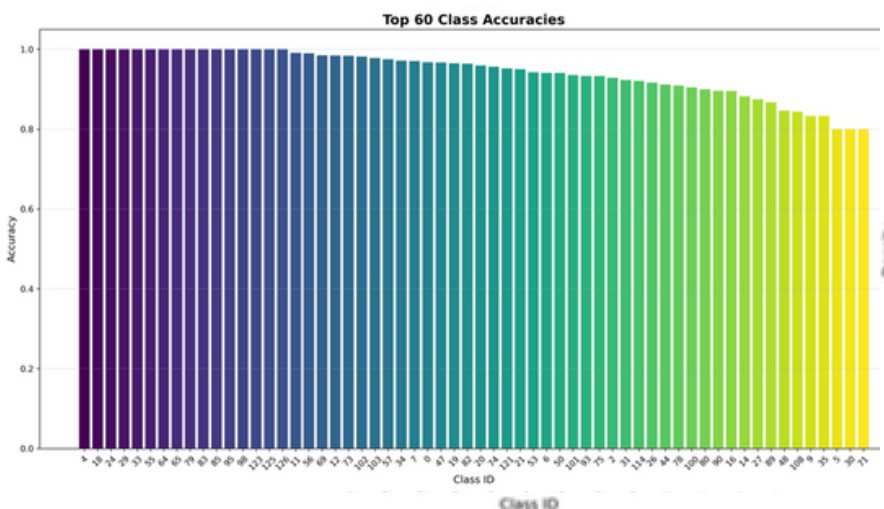


Business Analysis

Annual Operating Cost - Raspberry Pi Microscopy System (Total: ₹4,900 per year)



ML Model Outcomes



MODEL PERFORMANCE COMPREHENSIVE SUMMARY

OVERALL METRICS:

- Test Accuracy: 0.9563 (95.63%)
- Test Loss: 0.1836
- Total Test Samples: 10,000

CLASSIFICATION DETAILS:

- Classes in Test Set: 90
- Correct Predictions: 9,563
- Incorrect Predictions: 437

CONFIDENCE ANALYSIS:

- Mean Confidence: 0.934
- Correct Predictions Confidence: 0.947
- Incorrect Predictions Confidence: 0.659
- High Confidence (>0.8): 8,936 (89.4%)
- Low Confidence (<0.5): 235 (2.4%)

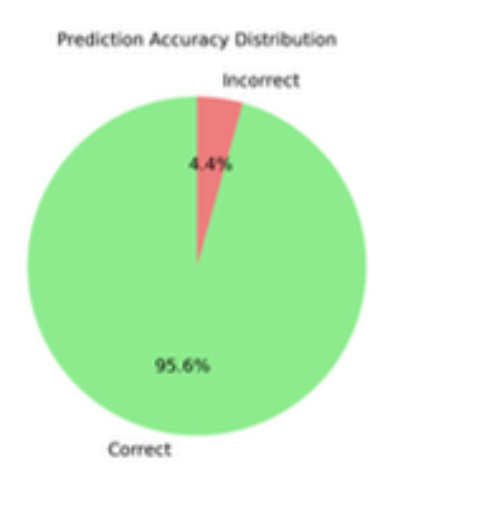
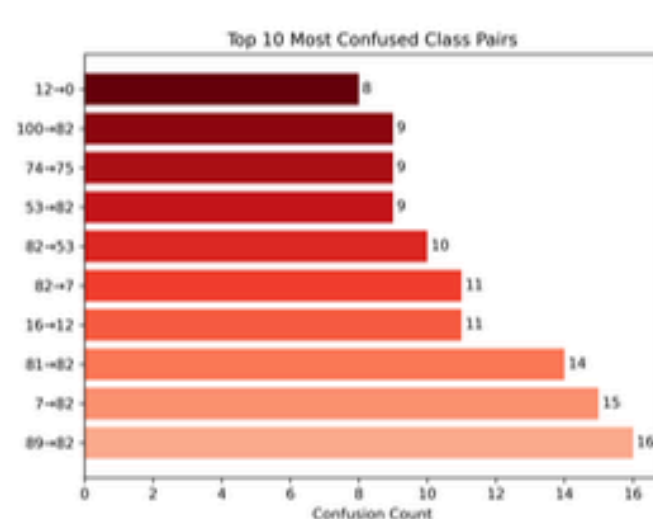
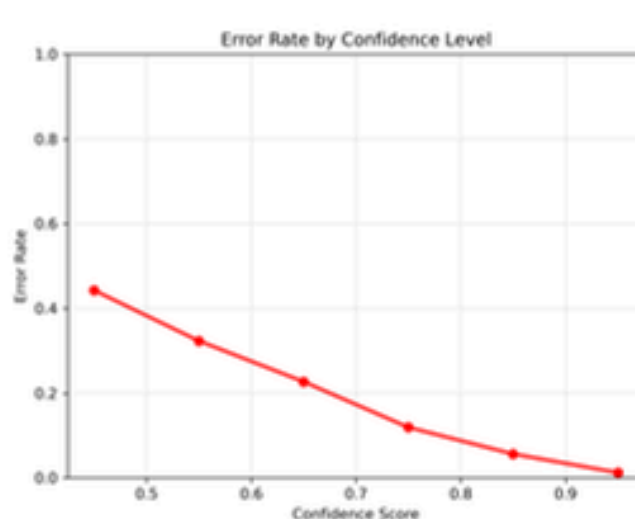
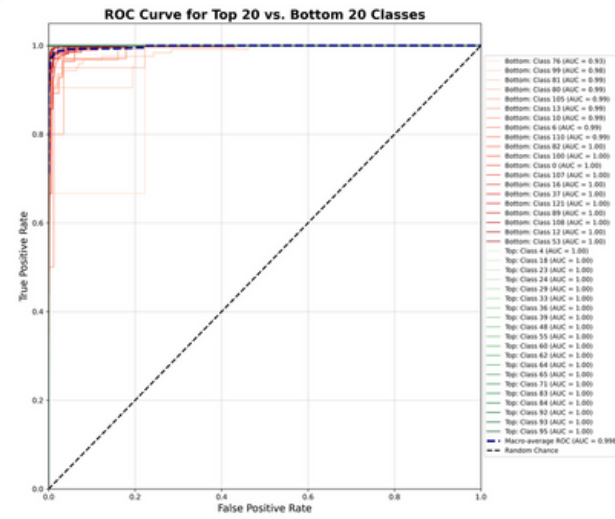
ADVANCED METRICS:

- ROC AUC (Macro Average): 0.9986
- ROC AUC (Weighted Average): 0.9980

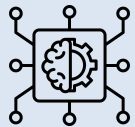
OVERALL ASSESSMENT:

- EXCELLENT - Ready for production deployment

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Scientific Impact



Hybrid AI Workflow with Dual model Pipeline

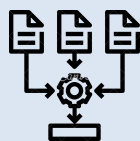


Continuous Learning Feedback Loop



Research-Grade Classification Accuracy

Operational Impact



Lightweight Quantized Model Inference

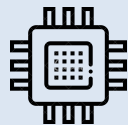


Adaptive Sampling Rate Control



Cloud-Enabled Manual Quality Check

Economical Benefit



Compact Hardware Footprint



Reduced Manual Taxonomist Workload



Low - Cost Large -Scale Rollout

Ecological Impact



Distributed Marine Monitoring Network



Climate-Driven Species Shift Tracking



Supports Sustainable Fisheries Decisions

Conclusion

A strong pipeline that keeps getting better is produced by combining remote training, on-device detection and counting, and cloud retraining

Complements automated processing by enabling specialists to verify predictions and improve models

System gets smarter as new species are added, without needing hardware replacement

Raspberry Pi platform offers portability, low power use, and compatibility with marine labs and vessels

Enables high-quality, continuous datasets for marine biodiversity research and long-term ecosystem management.

References

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